

Logical errors of the gross code: what they are, how to find them, how to check them

rl-for-qec notes

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These notes explain the ideas behind the coset catalogue in `src/train_utils/coset_catalogue.py` (exact, slow) and `coset_enumerate.py` (sampling, fast). Everything is linear algebra over $\mathbb{F}_2 = \{0, 1\}$ (addition is XOR). We treat X-type errors; Z-type errors are the mirror image.

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1 The lattice: a 6×12 torus with two qubits per site

The gross code $[[144, 12, 12]]$ lives on an $l \times m = 6 \times 12$ grid with periodic boundaries (a torus). A site is labelled (a, b) : $a \in \{0..5\}$ counts steps to the right (the x direction), $b \in \{0..11\}$ steps up (the y direction).¹ Each site carries four objects: an **L qubit**, an **R qubit**, an **X check** and a **Z check**. Hence $2 \cdot 72 = 144$ data qubits, 72 X checks and 72 Z checks. In the code, L qubit (a, b) has index $12a + b$ and R qubit (a, b) has index $72 + 12a + b$.

The natural picture is the one of the toric code (fig. 1): a square lattice with X checks on the vertices, Z checks on the faces, L qubits on the vertical edges and R qubits on the horizontal edges. Drawn this way every check touches its four nearest qubits, plus two distant ones.

¹Bravyi et al. write the gross code with $l = 12, m = 6$; our configuration has $l = 6, m = 12$ with the same A, B . Swapping $x \leftrightarrow y$ together with $L \leftrightarrow R$ maps one onto the other, so it is the same code.

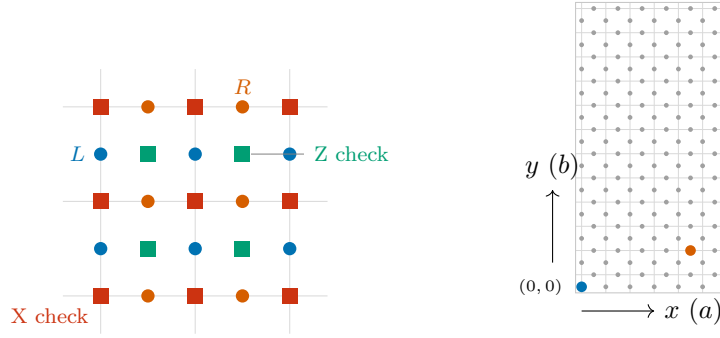


Figure 1: Left: a patch of the lattice. X checks (red) on vertices, Z checks (green) on faces, L qubits (blue) on vertical edges, R qubits (orange) on horizontal edges. Right: the whole 6×12 torus (grey dots: the 144 qubits; coloured: the L and R qubit of site $(0,0)$). Leaving on one side re-enters on the other.

2 What does $A = x^3 + y + y^2$ mean?

2.1 Monomials are shifts

Let x be “move one step right” ($a \rightarrow a + 1$) and y “move one step up” ($b \rightarrow b + 1$), both with wrap-around. As 72×72 matrices these are permutation matrices, $x = S_6 \otimes I_{12}$ and $y = I_6 \otimes S_{12}$ with S the cyclic shift. Then a monomial $x^i y^j$ is “move i right, j up”, and a *polynomial* is a *set of moves*: $x^3 + y + y^2$ means “three moves: 3 right; 1 up; 2 up”. The moves start at the *origin*: the qubit of the same site, the monomial 1. Products of monomials add the moves, and $x^6 = y^{12} = 1$ (going around the torus).

2.2 Checks are stencils

The check matrices are

$$H_X = [A \mid B], \quad H_Z = [B^\top \mid A^\top], \quad A = x^3 + y + y^2, \quad B = y^3 + x + x^2.$$

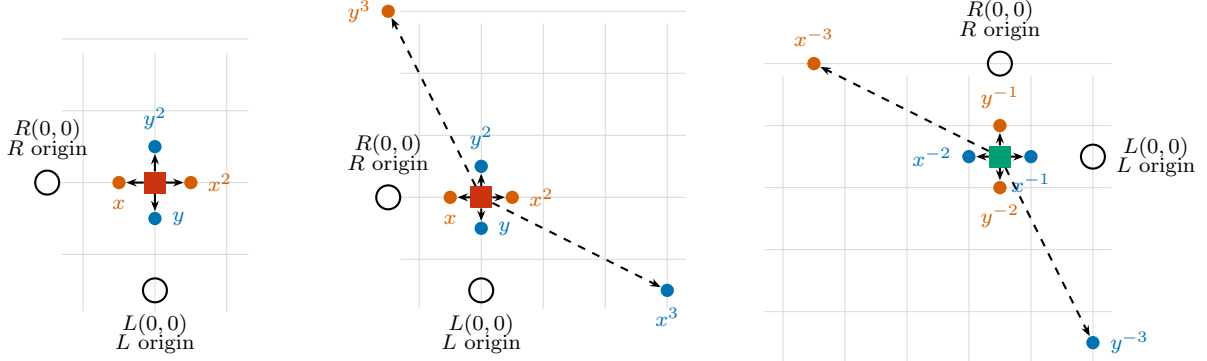
Read row (a, b) of H_X : the X check at site (a, b) acts on the L qubits reached from $L(a, b)$ by the moves in A , and on the R qubits reached from $R(a, b)$ by the moves in B . Each check is therefore the same six-legged *stencil*, copied to every site (fig. 2; the whole lattice with two stencils is fig. 3). The Z checks use the transposes, i.e. the *reversed* moves (x^{-1} = one step left, y^{-1} = one step down), with the roles of A and B swapped: B^{-1} on L , A^{-1} on R (fig. 2c).

Where is the origin of a check? Nothing has to be chosen. A check is labelled by its site (a, b) , and its moves always start from the two qubits with the *same* label: $L(a, b)$ for the moves on the L block, $R(a, b)$ for the moves on the R block. In the code there is no geometry at all: the X check of site (a, b) is row $12a + b$ of H_X , and a move $x^i y^j$ lands on column $12(a + i) + (b + j)$ (indices modulo 6 and 12) of the L block, or that plus 72 in the R block. Only in the *drawings* does a choice enter: each block is drawn with a fixed half-cell offset, chosen so that the four short legs of every check land on its four nearest edges. With that choice the origins always sit at the same place next to their check:

check	drawn on	L origin $L(a, b)$	R origin $R(a, b)$
X check	a vertex	1.5 steps down	1.5 steps left
Z check	a face	1.5 steps right	1.5 steps up

Rule of thumb: the origin is *one edge beyond the nearest short leg*. For an X check: one edge below its y leg (L) and one edge left of its x leg (R). For a Z check, whose moves are reversed: one edge

right of its x^{-1} leg (L) and one edge above its y^{-1} leg (R). Calling a different qubit the origin would multiply A or B by a fixed monomial; that only relabels the qubits (same code, shifted picture).



(a) Toric X check: $A = y + y^2$, $B = x + x^2$. (b) Gross X check: $A = (y + y^2) + x^3$, $B = (x + x^2) + y^3$. (c) Gross Z check: B^{-1} on L , A^{-1} on R (reversed moves).

Figure 2: The checks of site $(0,0)$ as stencils (X check: red square on a vertex; Z check: green square on a face). Black rings: the origins $L(0,0)$ and $R(0,0)$, i.e. the monomial 1; every arrow is a move *from the ring of its own colour block*: blue L qubits are reached from $L(0,0)$, orange R qubits from $R(0,0)$. E.g. in (b) x^3 is the L qubit three steps right of $L(0,0)$, y^3 the R qubit three steps up from $R(0,0)$; in (c) y^{-3} is the L qubit three steps down from $L(0,0)$. The origins sit 1.5 steps from their check: below and left of an X check, right of and above a Z check (table in the text).

Gross = toric + two long legs. Drop x^3 and y^3 and you are left with $A = y(1+y)$, $B = x(1+x)$. Up to a relabelling of qubits (the monomial factors y , x are just shifts), this is the toric code $A = 1 + y$, $B = 1 + x$, whose checks are the four nearest edges (fig. 2a). The gross code keeps the same idea, “one stencil copied everywhere”, and adds one long leg on each block. The long legs x^3 and y^3 are what makes the code non-local on a 2D chip, and also what buys $k = 12$ logical qubits and distance 12 from only 144 qubits (the toric code needs 288 qubits for $k = 2$, $d = 12$).

2.3 Why X and Z checks commute

A Z check and an X check must overlap on an even number of qubits. In matrix form, $H_X H_Z^T = AB + BA = 2AB = 0$, because shifts commute ($AB = BA$). The picture behind this (fig. 4): the X check at α reaches L qubit $\alpha + u$ for $u \in A$, and the Z check at β reaches L qubit $\beta - v$ for $v \in B$. They meet on L when $\beta - \alpha = u + v$. On R the X check moves by $v \in B$ and the Z check back by $u \in A$, so they meet when $\beta - \alpha = v + u$, the same condition. Every shared L qubit comes with a shared R qubit: “first u then v ” versus “first v then u ”.

3 Checking an error: three questions

Let $e \in \mathbb{F}_2^{144}$ be an X error, $e_q = 1$ if qubit q is flipped. In a decoder evaluation, e is the *residual*: true error plus the decoder’s correction.

1. **Is it detected?** The syndrome is $\sigma = H_Z e$: Z check j fires iff it overlaps e on an odd number of qubits. If $\sigma \neq 0$ the decoder sees the error.
2. **Is it harmless?** If $\sigma = 0$, the error is invisible. Invisible errors are either products of X checks (stabilisers, they do nothing to the encoded state) or logical operators (they do).

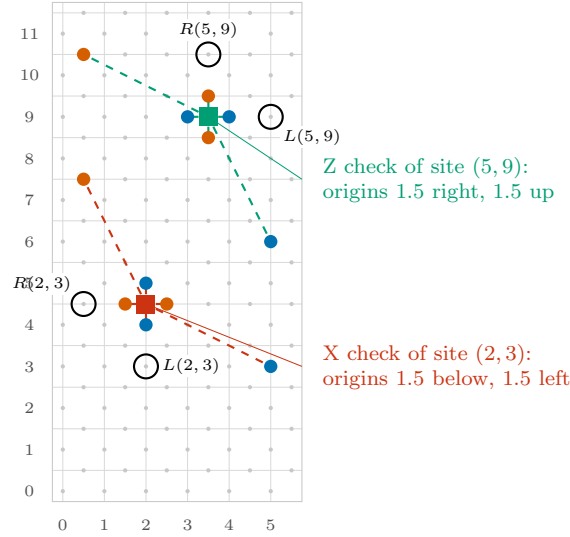


Figure 3: The whole gross code (x right, y up; tick labels are a and b) with two checks drawn. Each touches the four edges around it (solid) and two distant qubits (dashed); black rings mark the origins of its site, labelled $L(a,b)$ and $R(a,b)$: the X check of site $(2,3)$ moves from $L(2,3)$ and $R(2,3)$, the Z check of site $(5,9)$ from $L(5,9)$ and $R(5,9)$. The X check uses A from the L origin and B from the R origin; the Z check uses the reversed moves, B^{-1} on L and A^{-1} on R . Checks near the border have legs that wrap around the torus.

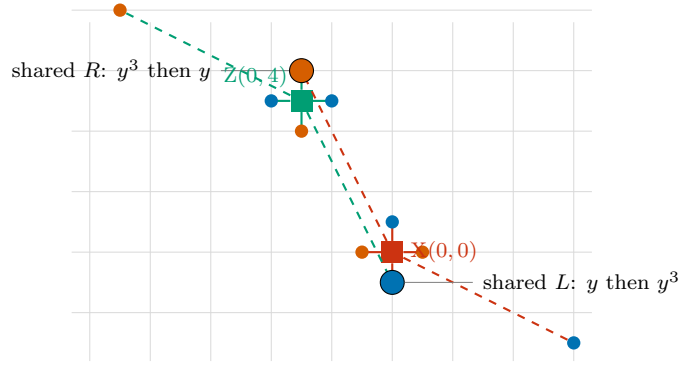


Figure 4: The X check at $(0,0)$ and the Z check at $(0,4)$ share exactly two qubits, one L and one R . Their offset $y^4 = y \cdot y^3$ can be split as “ $y \in A$ then $y^3 \in B$ ” (meeting on L : a local leg of X, a long leg of Z) or “ $y^3 \in B$ then $y \in A$ ” (meeting on R : a long leg of X, a local leg of Z).

Example: the stencil of one X check, used as an error, is invisible *and* harmless; it has weight 6.

3. **Which logical error?** Take L_Z , 12 Z-type logical operators (a basis of $\ker H_X$ modulo row H_Z). Each one is a “probe” that commutes with all checks; it anticommutes with e iff e flips that logical qubit. The *coset* of e is

$$s = L_Z e \in \mathbb{F}_2^{12}.$$

$s = 0$ means stabiliser; $s \neq 0$ is a logical error, and bit i of s says whether logical qubit i was flipped. We store s as the integer $c = \sum_i s_i 2^i$, the *coset id*.

Two invisible errors with the same s differ by a stabiliser, so they are the same error as far as the encoded information is concerned: a coset is a whole family of equivalent patterns. There are $2^{12} - 1 = 4095$ non-trivial X cosets. (The count of logical qubits: $k = n - \text{rank } H_X - \text{rank } H_Z = 144 - 66 - 66 = 12$.)

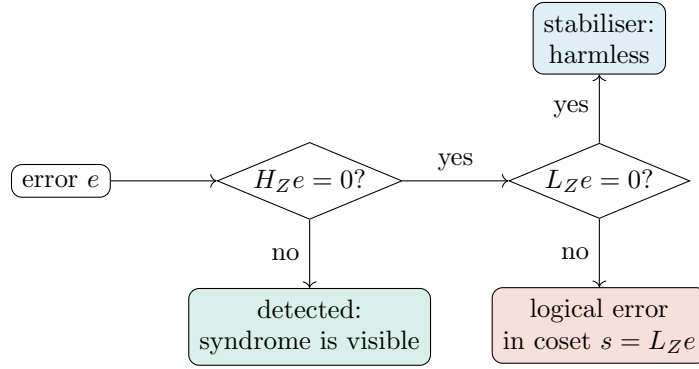


Figure 5: Classifying an X error. Both tests are one matrix–vector product mod 2.

4 An example: a weight-12 logical

The distance is 12: no invisible, non-trivial error is lighter. Figure 6 shows one that reaches this bound. It flips only L qubits, at

$$f(x, y) = (1 + y^6) (1 + x^2 + y^3(x + x^2 + x^3 + x^4)),$$

i.e. each monomial $x^a y^b$ of f is a flipped qubit at (a, b) .

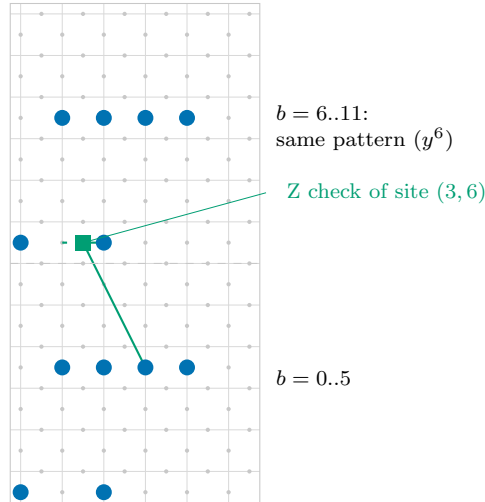


Figure 6: A weight-12 logical X error (coset id 128). Blue: flipped L qubits. Green: the three L legs of the Z check of site $(3, 6)$ (moves x^{-1}, x^{-2} to the edges left and right of it, y^{-3} far down; the dashed leg ends on an unflipped qubit). It overlaps the error on two qubits, so it does not fire. All 18 Z checks that touch the pattern overlap it on exactly two qubits, so the syndrome is zero.

Polynomial view. For an error f on L only, the Z checks see it through $B^\top$, which in polynomial language is multiplication by B . So

$$f \text{ on } L \text{ is invisible} \iff B(x, y) f(x, y) = 0 \quad \text{in } \mathbb{F}_2[x, y]/(x^6 - 1, y^{12} - 1).$$

This is a “null space” condition on a much smaller space (only 72 unknowns), and the null space has dimension 12: one can list *all* $2^{12} = 4096$ L -only invisible errors in a tenth of a second. Of these, 36 have weight 12, and they are exactly the 36 shifts of fig. 6 (see next section).

5 Orbits: the same pattern, shifted

Shifting every qubit by the same move $x^i y^j$ maps checks to checks (every check is the same stencil). So a shifted logical is still invisible and has the same weight and shape, but it is in general a *different coset*: it flips a different combination of the 12 logical qubits (fig. 7). All cosets obtained from one by shifting form an *orbit*.

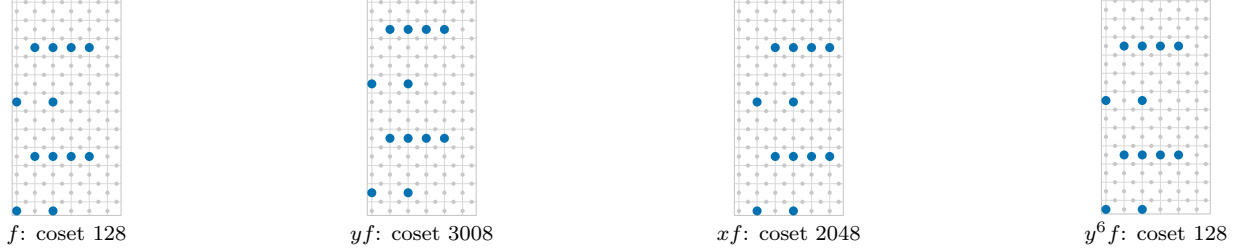


Figure 7: One orbit. Shifting the logical of fig. 6 by y or by x gives other cosets; shifting by y^6 gives the *same* pattern back, because $f = (1 + y^6)(\dots)$. Of the 72 shifts, pairs give the same coset, so this orbit has $72/2 = 36$ cosets.

In general the orbit size is $72/|\text{Stab}|$, with $|\text{Stab}|$ the number of shifts that leave the coset unchanged. For the gross code the 4095 cosets fall into **155 orbits** of sizes 3, 9, 12 and 36. One representative per orbit is enough to see every distinct shape. Computing the orbits needs no optimisation: take *any* representative of a coset (from a basis of X logicals), shift it 72 times, read off L_Z of each copy.

6 Finding the lightest representative of each orbit

Finding *some* invisible non-trivial error is a null-space computation. Finding the *lightest* one in a given coset is hard:

$$\min |e| \quad \text{s.t.} \quad H_Z e = 0, \quad L_Z e = s.$$

The full null space $\ker H_Z$ has dimension $144 - 66 = 78$, i.e. $2^{78} \approx 3 \cdot 10^{23}$ vectors, far too many to list, and the minimum-weight problem for linear codes is NP-hard in general. Three practical routes:

1. Single-block null spaces (instant, exact, partial). As in the previous section: errors on L only need $Bf = 0$, errors on R only need $Ag = 0$. Each null space has dimension 12 and is listed completely:

block	null space dim	weight-12 logicals	weight-16 logicals
L only	12	36 words $\rightarrow$ 36 cosets (one orbit)	9 words $\rightarrow$ 3 cosets
R only	12	36 words $\rightarrow$ 36 cosets (same orbit)	9 words $\rightarrow$ 3 cosets

Patterns that use both blocks are not covered.

2. Information-set sampling (fast, no proof). Let G (78×144) be a basis of $\ker H_Z$. A trial: shuffle the 144 columns, row-reduce G . Each row now has a single 1 on 78 “pivot” columns and is free elsewhere. Every row, and every sum of two rows, is an invisible error with at most two 1s on the pivots; keep the light ones with $L_Z e \neq 0$. A fixed pattern of weight w is caught in one trial with probability

$$P_w = \sum_{j=0}^2 \binom{78}{j} \binom{66}{w-j} / \binom{144}{w}, \quad P_{12} \approx 7.0 \cdot 10^{-3}, \quad P_{14} \approx 1.7 \cdot 10^{-3},$$

i.e. about 140 (resp. 600) trials per pattern, at a few milliseconds per trial. This gives many orbits quickly. It only gives upper bounds: not finding an orbit proves nothing.

3. Exact integer program (slow, certified). Write each mod-2 equation as an integer one with a slack, $r \cdot e - 2t = \text{rhs}$, and minimise $\sum_q e_q$ (HiGHS, zero optimality gap). Closing the gap is a proof of minimality. Easy orbits (weight 12) take about a minute; proving that a coset has *no* representative of weight ≤ 14 often takes more than 10 minutes. Use it to certify a few orbits, not to scan all 155.

7 What the lowest-weight logicals look like

A run of information-set sampling (20 000 trials, 2 workers, 4 minutes; weights up to 16) found representatives for 95 of the 155 orbits:

weight	orbits	cosets covered	status
12	9	246	minimal (distance bound)
14	26	888	upper bound
16	60	1701	upper bound
> 16?	60	1260	not seen

No new orbit appeared after the first ~ 1000 trials, while each weight-12 pattern is expected about every 140 trials; the 60 orbits never seen are very likely heavier than 16, but this is not a proof. Figure 8 draws one representative per orbit: all nine of weight 12 and nine of the 26 of weight 14, each shifted to a compact position.

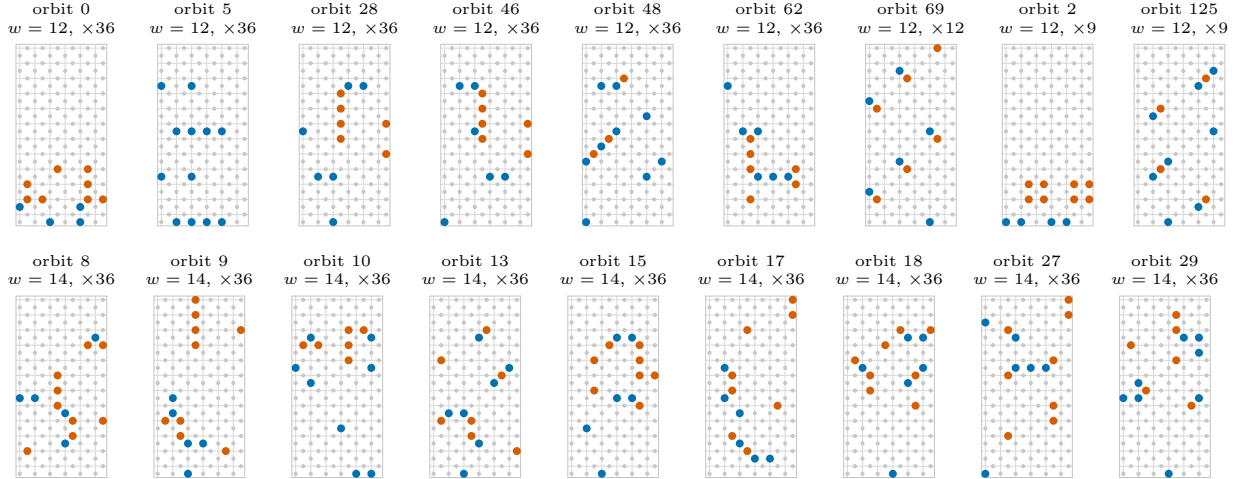


Figure 8: Lowest-weight logical X errors found, one panel per orbit (title: orbit id, weight w , number of cosets $\times N$ in the orbit). Top row: all orbits of weight 12; bottom row: nine of weight 14. Blue: flipped L qubits, orange: flipped R qubits; same layout as fig. 3 (x right, y up). An orbit with fewer than 72 cosets is a pattern that some shift maps to the same coset (like y^6 in fig. 7).

8 Checking a representative found by any method

1. $H_Z e = 0$: it is invisible;
2. $L_Z e = s \neq 0$: it is a logical error, in the claimed coset;
3. its weight $|e|$.

Minimality for free at weight 12. Since the distance is $d = 12$, every non-trivial invisible error has weight ≥ 12 . So *any* weight-12 representative, however it was found, is automatically the lightest in its coset. For heavier ones a proof needs the integer program (or another lower bound).

Status (2026-10-08)

orbit	smallest coset	cosets in orbit	weight	certified by
0	1	36	12	distance bound (and MILP)
2	7	9	12	distance bound (and MILP)
5	64	36	12	distance bound; single-block (L and R), fig. 6
1	3	12	14	MILP, proved optimal

Information-set sampling adds 6 more orbits of weight 12, 25 of weight 14 and 60 of weight 16 (fig. 8); data in `results/cosets/144_12_12_ldpc/enum_catalogue.csv`.